Consumption of meat and dairy and lymphoma risk in the European Prospective Investigation into Cancer and Nutrition.

The consumption of meat and other foods of animal origin is a risk factor for several types of cancer, but the results for lymphomas are inconclusive. Therefore, we examined these associations among 411,097 participants of the European Prospective Investigation into Cancer and Nutrition. During a median follow-up of 8.5 years, 1,334 lymphomas (1,267 non-Hodgkin lymphoma (NHL) and 67 Hodgkin lymphomas) were identified. Consumption of red and processed meat, poultry, milk and dairy products was assessed by dietary questionnaires. Cox proportional hazard regression was used to evaluate the association of the consumption of these food groups with lymphoma risk. Overall, the consumption of foods of animal origin was not associated with an increased risk of NHLS or HL, but the associations with specific subgroups of NHL entities were noted. A high intake of processed meat was associated with an increased risk of B-cell chronic lymphocytic leukemia (BCLL) [relative risk (RR) per 50 g intake = 1.31, 95% confidence interval (CI) 1.06-1.63], but a decreased risk of follicular lymphomas (FL) (RR = 0.58; CI 0.38-0.89). A high intake of poultry was related to an increased risk of B-cell lymphomas (RR = 1.22; CI 1.05-1.42 per 10 g intake), FL (RR = 1.65; CI 1.18-2.32) and BCLL (RR = 1.54; CI 1.18-2.01) in the continuous models. In conclusion, no consistent associations between red and processed meat consumption and lymphoma risk were observed, but we found that the consumption of poultry was related to an increased risk of B-cell lymphomas. Chance is a plausible explanation of the observed associations, which need to be confirmed in further studies.